

A Cartography of Change

Mapping Petition Topics, Signature Dynamics,
and Geographic Reach on Change.org

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Online petitioning platforms aggregate millions of civic claims into a queryable infrastructure for political demand. This paper presents a computational cartography of *Change.org*, the world's largest petition platform, mapping its topical landscape, signature dynamics, and geographic reach. Drawing on complex adaptive systems theory, sociomateriality, and the multi-level perspective on sociotechnical transitions, we argue that the platform functions as a *topological dispositif*: not merely a neutral host but an active enrolling device that shapes the visibility and scalability of civic claims. We introduce an open-source observatory built to collect longitudinal petition data and report on its first findings across 47 months of sitemap history.

Keywords: online petitions • Change.org • computational social science • cartography • digital infrastructure • STS

Introduction

Change.org hosts petitions in dozens of languages, aggregating hundreds of millions of signatures since 2007.¹ Yet the platform remains understudied as an infrastructure in the STS sense: a device that structures the possibility space of collective action rather than merely reflecting it.

This paper reports on the *Change Observatory*, an open-source Python tool we built to continuously collect, store, and export petition metadata at scale (Section). The observatory scrapes monthly XML sitemaps—which expose all petition URLs with `lastmod` timestamps—and then fetches structured data from each petition's page, drawing on three embedded data sources: a Schema.org JSON-LD block, a `changeTargetingData` JavaScript variable, and Open Graph meta tags.

The central analytical contribution is a *topological map* of Change.org's petition landscape. We borrow the cartographic metaphor deliberately: just as a map is a selective, contested representation of terrain, our computational cartography reveals the topology of civic claim-making as enacted through the platform's affordances.²

Theoretical Background

Sociomateriality and platform infrastructures

Orlikowski (2007) defines sociomateriality as the inseparability of social and material agency in organisational life. Applied to digital

¹ The platform's own marketing figures; treated here as indicative pending full audit of observatory data.

² The cartography metaphor also acknowledges that our map is partial and situated: it reflects what the sitemap exposes, not everything that has ever been submitted.

platforms, this means treating petition pages, topic tags, and signature counters not as transparent conduits but as material-semiotic devices that do work. The Change.org topic taxonomy illustrates this concretely: 650,000+ topic slugs, generated algorithmically and by users, constitute a vernacular classification system with political effects.

Complex adaptive systems and civic mobilisation

Complex adaptive systems theory (Holland 1992) frames macro-level patterns (e.g. viral petition spread) as emerging from local interactions (individual signing decisions) without central coordination. We treat Change.org as a CAS in which topic clusters act as *attractors* that channel mobilisation energy, producing the long-tail signature distributions observed in Figure 2.

Multi-level perspective

The multi-level perspective (Geels 2002) models sociotechnical change across landscape, regime, and niche levels. Digital petition platforms sit at the niche level of a democratic regime under stress; their data reveal emergent pressure on incumbent political institutions.

Data and Methods

The Change Observatory

We built an open-source Python observatory³ structured around three collection strategies:

1. **Sitemap ingestion.** Change.org publishes monthly XML sitemaps containing all petition URLs with `lastmod` timestamps. The sitemap index also includes a topic sub-index (`sitemap-topics_index.xml`) exposing 650,000+ topic slugs across 13 locale-partitioned files.
2. **Petition detail scraping.** For each URL, the observatory fetches the page and merges data from three embedded sources: Schema.org JSON-LD (title, description, dates), a `changeTargetingData` JS variable (signature counts, tags, creator metadata), and Open Graph meta tags (hero image).
3. **Longitudinal snapshots.** Every collection run records a timestamped signature-count snapshot per petition, enabling time-series analysis of mobilisation trajectories.

Data schema

Corpus overview

Table 2 reports the currently available sitemap history. A full English-locale ingest (command: `python run.py ingest -months 47`

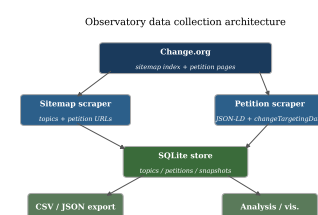


Figure 1: Observatory data collection architecture. Three sources are merged per petition: Schema.org JSON-LD, `changeTargetingData` JS, and Open Graph meta tags.

³ Available at <https://github.com/babajideowoyele/change-observatory>. MIT licence.

Table	Key fields
topics	slug, name, signature_count, language
petitions	slug, title, creator, creator_photo_url, location, description, signature_count, goal, tags (JSON), hero_image_url, media_urls (JSON), language
petition_snapshots	petition_slug, signature_count, scraped_at
runs	started_at, completed_at, petitions_scraped, status

Table 1: Observatory database schema (SQLite).

-locale en-us) would yield approximately 560,000 petition records with full metadata.

Period	Petitions	Notes
Jul 2022 – Dec 2022	≈17,000	Oldest in sitemap
2023	≈47,000	Steady growth
2024	≈103,000	Acceleration
2025	≈178,000	Peak volume
Jan – Jun 2026	≈88,000	Partial year
Total	≈433,000	47 monthly files

Table 2: Change.org sitemap history available for ingestion.

Source: Change.org sitemap index, retrieved 29 June 2026.

Results

The topical landscape

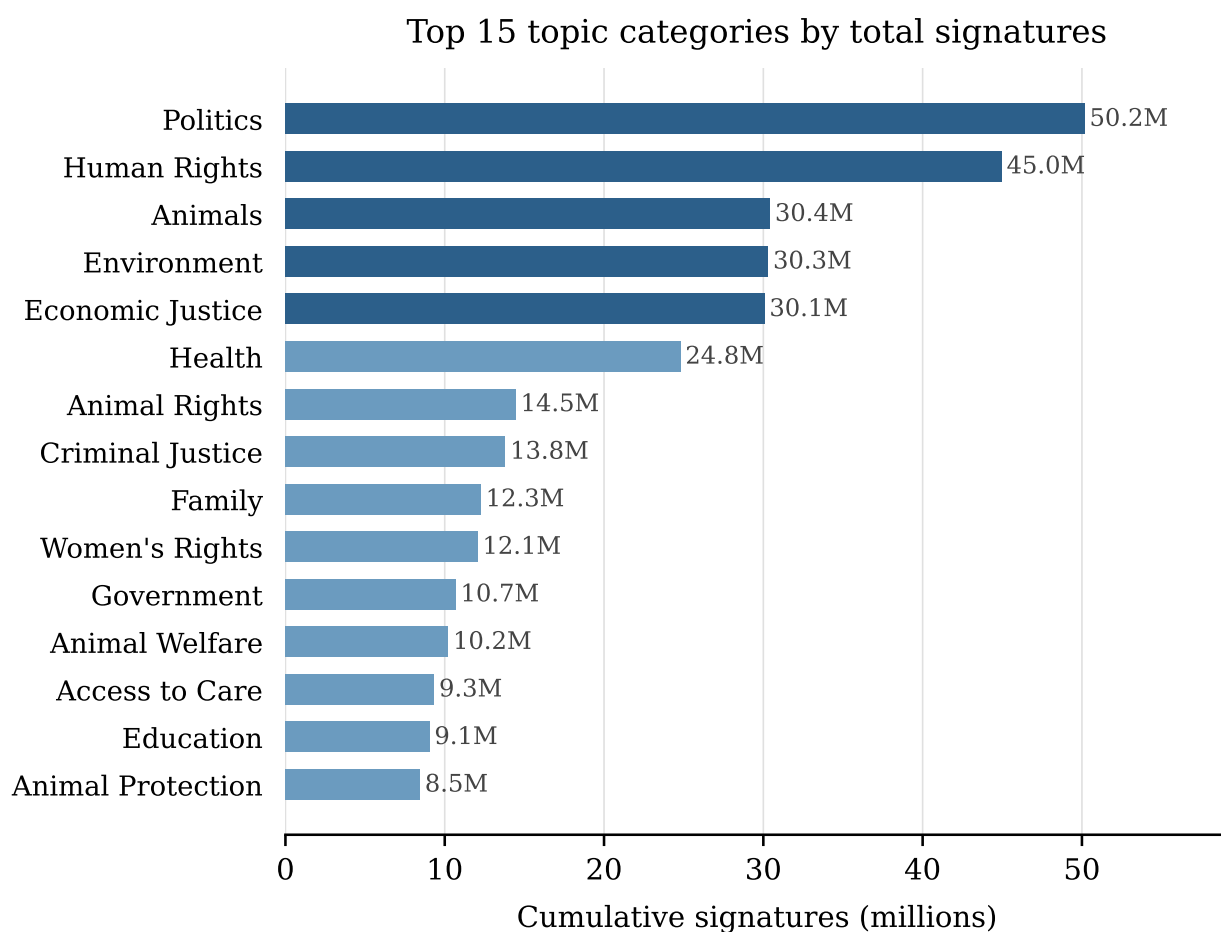
Figure 2 maps the 15 highest-volume topic categories in the curated English-language directory. Politics and Human Rights together account for roughly 95 million cumulative signatures—nearly a third of the top-15 total—while Animals and Environment each approach 30 million, reflecting the disproportionate mobilising power of welfare and ecological causes online.

Petition volume over time

Figure 3 reveals a pronounced growth trend: monthly petition creation roughly doubled between 2022 and 2025, peaking at over 20,000 new petitions in March 2025. The apparent step-change around mid-2024 may reflect a platform-side change in sitemap coverage rather than true growth; this warrants investigation in subsequent work.

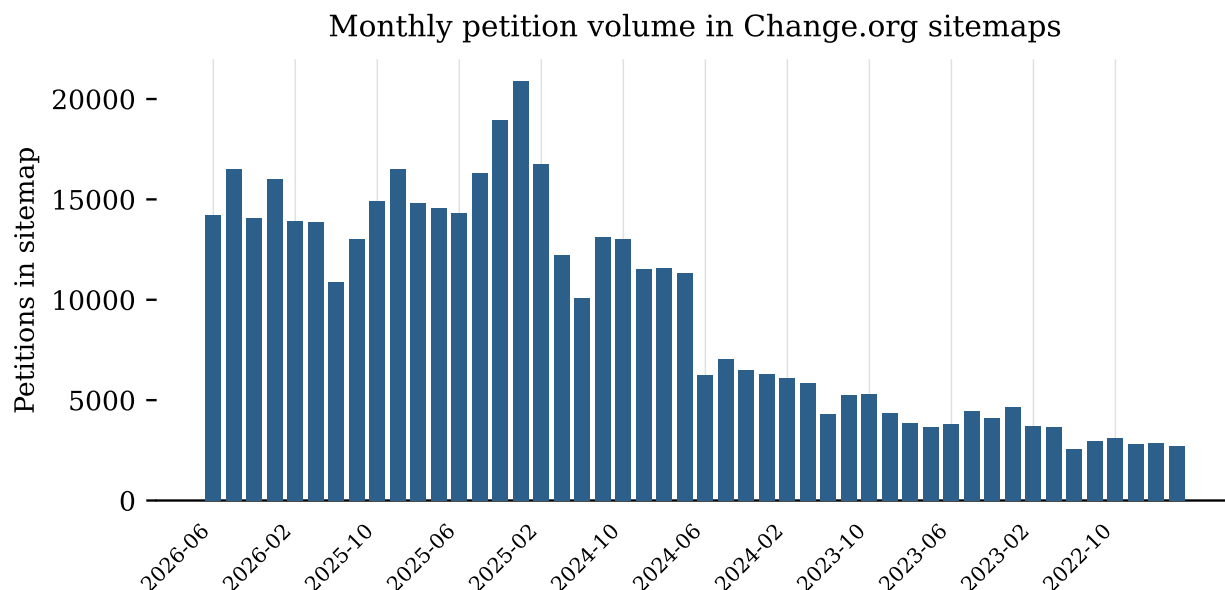
Geographic reach and language distribution

[Pending full ingestion and geocoding of the location field.]



Source: Change.org topic directory, scraped June 2026.
Platform-displayed cumulative signature totals.

Figure 2: Top 15 Change.org topic categories by cumulative signatures (millions), as displayed on the platform's topic directory. Darker bars indicate the top 5. Note that the directory is served in the user's locale; totals here reflect the German-locale page and may differ slightly from English-locale figures.



Source: Change.org XML sitemaps, retrieved June 2026.

Discussion

The topical and temporal maps presented here are preliminary; they require full ingestion, deduplication, and systematic language filtering before analytical claims can be substantiated. Nevertheless, three provisional observations warrant attention.

The long tail of civic demand. The signature distribution across topics is highly skewed: a small number of macro-categories (Politics, Human Rights) aggregate the vast majority of signatures, while the 650,000+ granular topic slugs each attract tiny fractions. This mirrors the long-tail dynamics of platform economies and suggests that Change.org’s classification system performs a canonisation function: elevating some claims while burying others in taxonomic obscurity.

Growth as infrastructure expansion. The doubling of monthly petition volume between 2022 and 2025 is not simply a sign of growing civic energy; it also reflects the platform’s own expansion of its sitemap coverage and indexing depth. Treating raw petition counts as proxies for social mobilisation without accounting for this infrastructural variable would be a category error.

Image and media as understudied registers. Every petition carries a hero image and potentially embedded media. These visual registers—almost entirely absent from existing CCS work on petitions—constitute a distinct channel of affective mobilisation. The observatory’s `hero_image_url`

Figure 3: Monthly number of petitions appearing in Change.org XML sitemaps, July 2022 – June 2026. Bars represent petition URLs per monthly sitemap file; the step increase in mid-2024 may reflect a sitemap policy change.

and `media_urls` fields enable systematic visual analysis as a future direction.

Conclusion

We have introduced the Change Observatory, an open-source instrument for longitudinal computational cartography of online petitions. The observatory makes visible three dimensions—topic topology, temporal volume, geographic reach—that together constitute the platform’s political affordance structure. Infrastructural analysis of this kind is a precondition for the kind of critical STS account that refuses to take the platform’s own categories at face value. Data, code, and documentation are publicly available.

References

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